

Name: M Moazam FA23-BSE-071

Q#02: Case Study

Online Food Delivery System

Use Case

Apply Discount Coupon

Classes

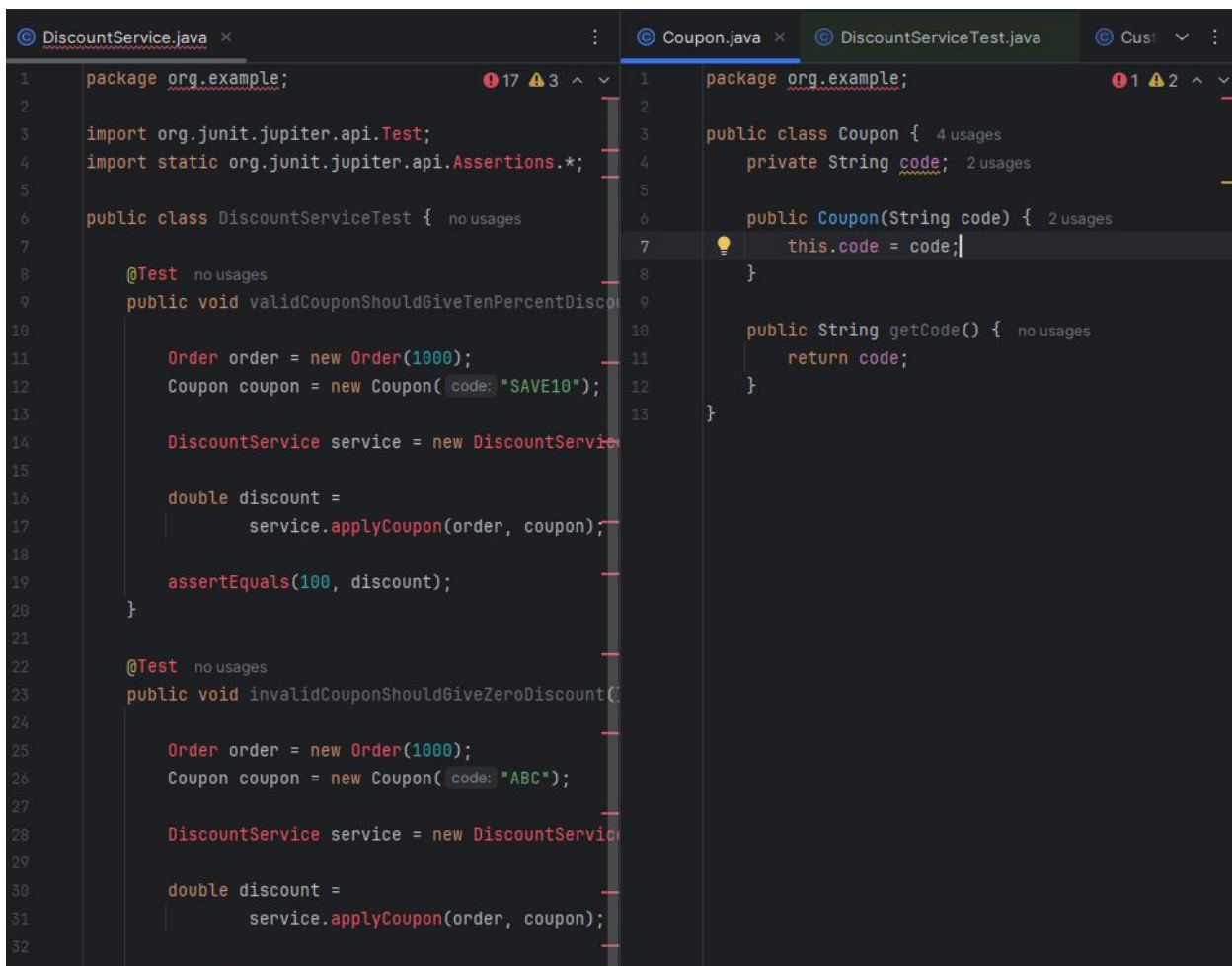
1. Customer
2. Order
3. Coupon
4. DiscountService

Core Class for TDD

DiscountService

TDD Cycle

1. Write test first (RED)
2. Implement DiscountService (GREEN)
3. Refactor if needed (REFACTOR)



```
DiscountServiceTest.java
package org.example;

import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

public class DiscountServiceTest {

    @Test
    public void validCouponShouldGiveTenPercentDiscount() {

        Order order = new Order(1000);
        Coupon coupon = new Coupon("code: SAVE10");

        DiscountService service = new DiscountService();

        double discount =
            service.applyCoupon(order, coupon);

        assertEquals(100, discount);
    }

    @Test
    public void invalidCouponShouldGiveZeroDiscount() {

        Order order = new Order(1000);
        Coupon coupon = new Coupon("code: ABC");

        DiscountService service = new DiscountService();

        double discount =
            service.applyCoupon(order, coupon);
    }
}

Coupon.java
package org.example;

public class Coupon {
    private String code;

    public Coupon(String code) {
        this.code = code;
    }

    public String getCode() {
        return code;
    }
}
```

```
1 package org.example;
2
3 public class Coupon { 4 usages
4     private String code; 2 usages
5
6     public Coupon(String code) { 2 usages
7         this.code = code;
8     }
9
10    public String getCode() { no usages
11        return code;
12    }
13 }
```

```
1 package org.example;
2
3 public class Customer { no usages
4     private String name; 2 usages
5
6     public Customer(String name) { no usages
7         this.name = name;
8     }
9
10    public String getName() { no usages
11        return name;
12    }
13 }
```

```
1 package org.example;
2
3 public class Order { no usages
4     private double totalAmount; 2 usages
5
6     public Order(double totalAmount) { no usages
7         this.totalAmount = totalAmount;
8     }
9
10    public double getTotalAmount() { no usages
11        return totalAmount;
12    }
13 }
```

Q#01:

```
1  import org.junit.jupiter.api.Test;
2  import static org.junit.jupiter.api.Assertions.*;
3
4  public class PalindromeCheckerTest {
5
6      @Test
7      public void positiveTestPalindrome() {
8          PalindromeChecker checker = new PalindromeChecker();
9
10         boolean result = checker.isPalindrome("madam");
11
12         assertTrue(result);
13     }
14
15     @Test
16     public void negativeTestNotPalindrome() {
17         PalindromeChecker checker = new PalindromeChecker();
18
19         boolean result = checker.isPalindrome("civic");
20
21         assertFalse(result);
22     }
23 }
```

```
1 public class PalindromeChecker { 4 usages
2
3     public boolean isPalindrome(String text) { 2 usages
4         String reversed = new StringBuilder(text).reverse().toString();
5         return text.equalsIgnoreCase(reversed);
6     }
7 }
```